

CSCE 421: Machine Learning (Spring 2025)

Homework #6

Due 5/2/2025, 11:59 pm

1. We will NOT accept late submissions for this assignment due to tight timing on final grades.
 2. You need to submit (1) a report in PDF and (2) your code files, both to Canvas.
 3. Your PDF report should include (1) answers to the non-programming part, and (2) results and analysis of the programming part. For the programming part, your PDF report should at least include the results you obtained, for example the accuracy, training curves, parameters, etc. You should also analyze your results as needed.
 4. Please name your PDF report “HW#_FirstName_LastName.pdf”. Please put all code files into a compressed file named “HW#_FirstName_LastName.zip”. Please submit two files (.pdf and .zip) to Canvas (i.e., do not include the PDF file into the ZIP file).
 5. Only write your code between the following lines. Do not modify other parts.
YOUR CODE HERE
END YOUR CODE
 6. All students are highly encouraged to typeset their reports using Word or L^AT_EX. In case you decide to hand-write, please make sure your answers are clearly readable in scanned PDF.
 7. Unlimited number of submissions are allowed and the latest one will be timed and graded. If you make a resubmission after the due date, it will be considered late.
 8. Please read and follow submission instructions. No exception will be made to accommodate incorrectly submitted files/reports.
 9. Please start your submission to Canvas at least 15-30 minutes before the deadline, as there might be latency. We do NOT accept E-mail submissions.
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1. (20 points) Let $A = U\Sigma V^T$ be the SVD of A , where $A \in \mathbb{R}^{m \times n}$, $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices, $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)$, and $r = \text{rank}(A)$. Show that
 - (a) The first r columns of U are eigenvectors of AA^T corresponding to nonzero eigenvalues.
 - (b) The first r columns of V are eigenvectors of $A^T A$ corresponding to nonzero eigenvalues.
2. (20 points) Given a symmetric matrix $A \in \mathbb{R}^{3 \times 3}$, suppose its eigen-decomposition can be written as

$$A = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{21} & u_{31} \\ u_{12} & u_{22} & u_{32} \\ u_{13} & u_{23} & u_{33} \end{pmatrix}. \quad (1)$$

What is the singular value decomposition of this matrix?

3. (60 points) **Principal Component Analysis:** In this homework, you will apply the principal component analysis to a collection of handwritten digit images from the USPS dataset. The USPS dataset is in the “data” folder: USPS.mat. The starting code is in the “code” folder. The whole data has already been loaded into the matrix A . The matrix A has shape 3000×256 and contains all the images. Each row in A corresponds to a handwritten digit image (between 0 and 9) with size 16×16 . You are expected to implement your solution based on the given codes. The only file you need to modify is the “solution.py” file. You can test your solution by running the “main.py” file.
- (a) (20 points) Complete the `_do_pca()` method. Your code will be tested on $p = 10, 50, 100, 200$, a total of four different numbers of the principal components.
 - (b) (10 points) Complete the `reconstruction()` method to reconstruct the reduced data.
 - (c) (10 points) Complete the `reconstruct_error()` function to measuring the reconstruction error.
 - (d) (20 points) Run “main.py” to see the reconstruction results and summarize your observations from the results into a short report. When you run the “main.py” file, a subset (the first two) of the reconstructed images based on $p = 10, 50, 100, 200$ principal components will be automatically saved on the “code” folder. Please attach these images to your report.

Note: You are NOT supposed to use existing PCA libraries; instead, you should write your own PCA. Please read the “Readme.txt” file carefully before you start this assignment.